

Next-generation prediction metrics for composite-based PLS-SEM

Next-generation
prediction
metrics for
PLS-SEM

Joe F. Hair Jr

Marketing, University of South Alabama, Mobile, Alabama, USA

5

Abstract

Purpose – The purpose of this study is to provide an overview of emerging prediction assessment tools for composite-based PLS-SEM, particularly proposed out-of-sample prediction methodologies.

Design/methodology/approach – A review of recently developed out-of-sample prediction assessment tools for composite-based PLS-SEM that will expand the skills of researchers and inform them on new methodologies for improving evaluation of theoretical models. Recently developed and proposed cross-validation approaches for model comparisons and benchmarking are reviewed and evaluated.

Findings – The results summarize next-generation prediction metrics that will substantially improve researchers' ability to assess and report the extent to which their theoretical models provide meaningful predictions. Improved prediction assessment metrics are essential to justify (practical) implications and recommendations developed on the basis of theoretical model estimation results.

Originality/value – The paper provides an overview of recently developed and proposed out-of-sample prediction metrics for composite-based PLS-SEM that will enhance the ability of researchers to demonstrate generalization of their findings from sample data to the population.

Keywords Benchmarking, Composite-based SEM, Cross-validation, Partial least squares, PLS-SEM, Out-of-sample prediction, Structural equation modeling

Paper type Research paper

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Introduction

Structural equation modeling (SEM) is an essential analytical tool in social sciences research, including information systems. Since its introduction composite-based PLS-SEM has been viewed as serving two important analytical roles – one is explanation and the other is prediction (Wold 1982; Jöreskog and Wold, 1982). This dual possibility is different from covariance-based SEM (Jöreskog, 1978; Rigdon, 1998; Diamantopoulos and Siguaw, 2006), which was designed only for explanatory purposes (Sarstedt *et al.*, 2016).

A primary objective of PLS-SEM is maximizing the amount of variance explained in the endogenous constructs of the structural model (often viewed as prediction) and at the same time facilitating explanation of the model relationships (Hair *et al.*, 2017a; Hwang *et al.*, 2020; Sarstedt *et al.*, 2017). Gregor (2006) refers to this characteristic as a combination of theory explanation and prediction that enables both prediction and the ability to describe the relationships among the theoretical constructs. In the past, researchers applying PLS-SEM have often emphasized the predictive nature of their analyses. Most researchers have interpreted the coefficient of determination (R^2), which assesses the in-sample model fit of the endogenous constructs' composite scores by using the model estimates to predict the individual case values of the total sample. But the R^2 value only assesses a model's explanatory power. It does not assess the model's out-of-sample predictive power with regard to predicting the values of new cases not included in the estimation process. Focusing on metrics that evaluate a model's explanatory power is problematic because an optimal predictive model may differ from an approach designed to identify the best explanatory model. In other words, a model designed in an explanatory context that exhibits good fit may perform poorly in terms of out-of-sample predictive power (Shmueli, 2010), thus limiting its practical usefulness.

Assessing a model's out-of-sample predictive power involves estimating the model on an initial analysis (training) sample and evaluating its predictive performance for a second set of



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data other than the original analysis (training) sample (Cepeda Carrión *et al.*, 2016; Hair *et al.*, 2019a; Shmueli *et al.*, 2016). A frequently used metric in this context is the Q^2 value (Geisser, 1974; Stone, 1974), which results from the blindfolding procedure (Chin, 1998). But the Q^2 procedure also does not utilize holdout samples. Most importantly, it does not clearly indicate whether the model has either good explanatory fit (as does R^2) or satisfactory predictive power (Shmueli *et al.*, 2019; Sarstedt *et al.*, 2017).

In light of these limitations, Shmueli *et al.*'s (2016) $PLS_{predict}$ proposed a holdout sample-based procedure that calculates case-level predictions at both an item and a construct level. The focus of the $PLS_{predict}$ metric, therefore, is to provide the ability to assess out-of-sample prediction for a single theoretical model in addition to in-sample prediction. More recently, several other alternatives have been developed that provide additional approaches to evaluate prediction when using composite-based PLS-SEM. In the following, I briefly summarize these next-generation prediction metrics and review choices researchers need to make when applying the procedures.

Out-of-sample prediction using $PLS_{predict}$

For many years, researchers have proposed the holdout sample approach to assess out-of-sample predictive power as an option with multivariate methods (Hair *et al.*, 1984), but it has very seldom been reported since software was not available to execute the process. To facilitate this type of analysis in PLS-SEM, Shmueli *et al.* (2016) proposed the $PLS_{predict}$ procedure. $PLS_{predict}$ applies a holdout sample approach to generate observation-level predictions on an item level (Shmueli *et al.*, 2019). Evaluation metrics such as R^2 and Q^2 reported for PLS-SEM results provide in-sample predictions, but $PLS_{predict}$ uses multiple randomly selected holdout samples to calculate out-of-sample predictive power. Information systems researchers using PLS-SEM should routinely report the results for the $PLS_{predict}$ procedure (Chin, 1998; Nitzland Chin, 2017).

Cross-validated prediction alternatives for PLS-SEM

A more recent advancement in the prediction-oriented assessment of PLS-SEM results is the cross-validated predictive ability test (CVPAT; Liengaard *et al.*, 2020). CVPAT is a statistical testing procedure that makes it possible to assess whether a proposed or alternative theoretical model achieves significantly better out-of-sample predictive power than another established model. Different from information theoretic model selection criteria whose objective is balancing model fit and predictive power in the PLS-SEM context (Danks *et al.*, 2020; Sharma *et al.*, 2020a, b), CVPAT focuses exclusively on a models' predictive power. Information systems researchers can use the CVPAT, therefore, for developing and validating theories that offer both explanation and prediction, as well as for comparing and selecting a model on which to base managerial and policy decisions. An example of when such a predictive model comparison would have been useful is the comparison by Mathieson (1991) of the technology acceptance model and the theory of planned behavior in terms of their ability to explain individuals' intentions to use information technologies.

To quantify the out-of-sample prediction errors, CVPAT relies on multiple cross-validation procedures (Stone, 1974) previously used in machine-learning prediction methods but only recently added for PLS-SEM (Shmueli *et al.*, 2016). Cross-validation assesses a model's out-of-sample predictive power (generalization error) by dividing the total sample into training and holdout sets multiple times (Hastie *et al.*, 2009). CVPAT-based cross-validation makes the assessment of a model's generalization performance possible, therefore, and reduces the likelihood of overfitting and underfitting. The test can also be applied when research involves complex models tested with relatively small data sets – as is frequently the

situation when using PLS-SEM. The CVPAT process (Lienggaard *et al.*, 2020) is shown in Table 1.

Application of the CVPAT procedure can be illustrated with the American Customer Satisfaction Index (ACSI), a frequently applied model in business research (Lienggaard *et al.*, 2020). Step 1 of the CVPAT process, therefore, is the decision to examine the relationship between customer satisfaction and customer loyalty or repurchase intentions (Fornell *et al.*, 1996; Anderson and Fornell, 2000). Figure 1 displays two models that differ based on whether satisfaction precedes perceived value, or vice versa. Both models have five constructs: perceived quality (PERQ), customer expectations (CUEX), perceived value (PERV), customer satisfaction (CUSA), customer loyalty (CUSL). One of the models is an established theoretical model (EM) and the other is an alternative model (AM). Most customer research supports the “value first” conceptualization of the EM where PERV serves as the immediate antecedent of CUSA (Hult *et al.*, 2019). But an alternative “satisfaction-first” conceptualization from the customer value equity literature proposes a model where PERV mediates the relationship between CUSA and CUSL (Johnson *et al.*, 2001).

The second step is to obtain solutions for the EM and AM with empirical data using PLS-SEM. The goal would be to determine whether the AM offers significantly higher out-of-sample predictive power than the EM. Step 3 is to execute the CVPAT procedure to facilitate comparison of the predictive models. Step 4 focuses on the final decision-making and model selection. Comparison of the results of the two theoretical models would enable the researcher to select either the EM or AM based on the model that exhibits the higher predictive accuracy.

Recent methodological research is extending CVPAT’s capabilities in two ways: (1) to include a stand-alone comparison of models with a naïve indicator-average benchmark, and (2) to assess theoretical models with regard to specific constructs, rather than evaluating all endogenous constructs simultaneously (Sharma *et al.*, 2020b). I am confident future research will advance the CVPAT method to further increase its scope of application in various model constellations such as mediation models (Sarstedt *et al.*, 2020).

Observations and conclusions

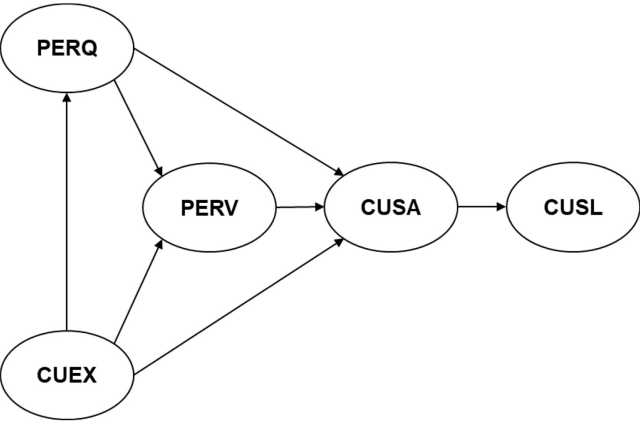
Assessing a statistical model’s predictive power is an essential element of almost all social sciences research. The composite-based PLS-SEM method facilitates both explanatory and predictive analyses in path models (Cheah *et al.*, 2020, Hair *et al.*, 2019b, c). In the past information systems researchers applying PLS-SEM have most often reported explanatory model assessments using in-sample metrics such as the R^2 and Q^2 (Hair *et al.*, 2017b; Lienggaard *et al.*, 2020). Forward-looking information systems researchers need techniques that complement their explanatory research predictive procedures and establish a stronger prediction orientation (Hofman *et al.*, 2017). Chin *et al.* (2020) conclude that CVPAT and PLS_{predict} are particularly well suited for executing out-of-sample prediction-oriented assessments and comparisons.

The cross-validation-based predictive methods described in this paper should be applied in the structural model assessment stage of confirmatory composite analysis (CCA) as discussed in Hair *et al.* (2019a, 2020). A description of the evolution of the CCA process over

Step 1	Examine theory and prior research of the established model (EM) and the alternative model (AM)
Step 2	Execute PLS-SEM analysis
Step 3	Apply CVPAT for predictive model comparison
Step 4	Make decision on model selection

Table 1.
Prediction-oriented
model selection
procedure based
on CVPAT

Established Model (EM)



Alternative Model (AM)

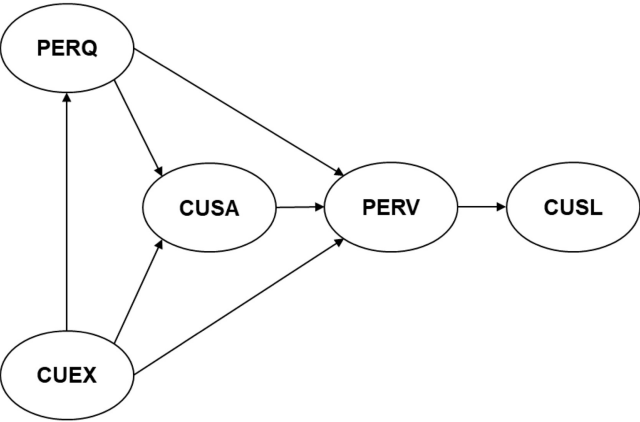


Figure 1.
Comparison of two
theoretical customer
loyalty models

the past 50 years is given in [Hair et al. \(2020\)](#). Application of the prediction assessment procedures described herein will increase confidence in the findings of information systems researchers and will be a substantial step in further improving prediction procedures for testing theoretical models.

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Further reading

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About the author

Joe F. Hair Jr is Director of the PhD Program and Cleverdon Chair of Business, Mitchell College of Business, University of South Alabama, USA. In 2018 and 2019, Joe was recognized by Clarivate Analytics for being in the top 1% globally of all Business and Economics professors based on his citations and scholarly accomplishments, which for his career exceed 231,000. He has authored over 75 editions of his books, including *Multivariate Data Analysis*, Cengage Learning, UK, 8th edition 2019 (cited 133,000+ times and one of the top five all-time social sciences research methods textbooks); *MKTG*, Cengage Learning, 13th edition, 2021; *Essentials of Business Research Methods*, Routledge, 4th edition 2020; *Essentials of Marketing Research*, McGraw-Hill, 5th edition 2020; and *A Primer on Partial Least Squares Structural Equation Modeling*, Sage, 3rd edition 2021. He also has published numerous articles in scholarly journals such as the *Journal of Marketing Research*, *Journal of Academy of Marketing Science*, *European Journal of Marketing*, *Organizational Research Methods*, *Journal of Advertising Research*, *Journal of Long Range Planning*, *Decision Sciences Journal*, *European Management Journal*, *Industrial Marketing Management*, *Journal of Retailing* and others. He has a new book on *Marketing Analytics*, forthcoming in 2021 (McGraw-Hill). Joe F. Hair Jr can be contacted at: jhair@southalabama.edu

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